

AUTOMATED FABRIC DEFECT IDENTIFICATION SYSTEM FOR TEXTILE MILLS

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Objective: Annotate fabric defect images across all 20 defect types in Roboflow, prepare the dataset for model training, and validate annotation quality through a defined QA workflow.

Building on the architecture and dataset selection from Milestone 1, Milestone 2 focused entirely on annotating the 846-image fabric defect dataset. The team manually labelled defects across all 20 categories using two strategies — bounding boxes for compact localised defects and smart polygons for irregular or boundary-following defects.

The annotation phase is the most critical step in the pipeline, as label quality and consistency directly determines YOLOv11 detection accuracy.

2. PROJECT MILESTONES – PROGRESS UPDATE

Milestone	Description	Status
M1	Architecture Design: Camera & lighting specs, annotation taxonomy, pipeline design, Power BI model	Done
M2	Annotation & Dataset Preparation: Annotate 846 images, train YOLOv11, evaluate precision/recall	Done
M3	Inference Pipeline: Python script for live scan, defect classification, meter positions, batch logs	Upcoming
M4	Power BI Dashboard: Defect distribution, batch scores, shift trends, rejection risk	Upcoming
M5	Quality Improvement Analysis: Benchmark vs. manual inspection; estimate export penalty reduction	Upcoming

3. PROBLEM STATEMENT & CHALLENGES

3.1 Core Problem

Textile lines operate at high speed across diverse patterns, textures, and colours. Manual inspection fails due to:

- Human fatigue over extended inspection shifts

- Inconsistent defect recognition across operators
- Production speed making real-time manual inspection infeasible
- Visual similarity between defect types causing frequent misclassification

3.2 Annotation Challenges

Challenge	Impact
Limited Indian fabric datasets	Few reference images for rare defect classes
Pattern/texture variation	Defect boundaries harder to distinguish from background weave
Low image resolution	Small defects near 10×10 px threshold; some skipped
Complex backgrounds	False boundary placement risk; tight-fit discipline required
High inter-class similarity	E.g. loose thread vs. snag; required side-by-side reference
Motion blur	Blurred edges made polygon placement less precise
Rare defect classes	Low sample count for press-off, seam puckering; imbalance risk

4. PROJECT GOALS ADDRESSED

1. **Diverse annotated dataset** — All 846 images annotated across 20 categories covering varied fabric patterns, colours, and textures.
2. **Accurate bounding boxes** — Tight-fit annotation enforced; maximum 5 px padding; 10×10 px minimum box size.
3. **Classification of similar defects** — Annotators used side-by-side defect reference guides to distinguish loose-thread, snag, and mis-threaded.
4. **Rare defect handling** — Multi-label annotation per image; rare class images flagged for augmen-

tation in Milestone 3.

5. **Scalable QA workflow** — Four-step QA pipeline with dual annotator review and IoU validation implemented in Roboflow.

5. ROBOFLOW PROJECT SETUP

5.1 Project Configuration

Parameter	Value
Platform	Roboflow Universe
Project Type	Object Detection
Total Images	846
Total Classes	20
Export Format	YOLOv11 (PyTorch)

Setup steps: (1) Created Object Detection project; (2) Added all 20 class labels from taxonomy; (3) Uploaded 846 images by forking from Roboflow Universe; (4) Distributed images to annotation queues across team.

5.2 Defect Class Reference

#	Class Name	#	Class Name
1	Abrasion-marks	11	Miss-Pick
2	Crease	12	Open-Seam
3	Damaged-Selvedge	13	Piling
4	Fly-Yarn	14	Press-Off
5	Hole	15	Seam-Puckering
6	Knots	16	Skip-Stitch
7	Loose-Thread	17	Slub
8	Looseness	18	Snag
9	Misplaced/Mis-Threaded	19	Spot
10	Misprinting	20	Weft-Bar/Warp-Bar

6. ANNOTATION STRATEGY

6.1 Annotation Type Per Defect

Bounding Box — compact, localised defects with well-defined rectangular extent:

hole · slub · knots · piling · spot · fly yarn · miss pick · weft-bar/warp-bar · press-off · looseness · misprinting · abrasion marks

Tight box drawn around defect; max 5 px padding; maps directly to YOLO format (*xcenter, ycenter, width, height*) normalised to image dimensions.

Smart Polygon — irregular, elongated, or boundary-following defects:

crease · open seam · seam puckering · damaged selvedge · snag · skip stitch · loose thread · misplaced/mis-threaded

Polygon vertices placed along defect boundary.

Roboflow’s ”Find Objects with AI” tool used for initial placement followed by manual adjustment.

6.2 Annotation Rules

Rule	Specification
Tight Fit	Box/polygon hugs defect edge; max 5 px padding
Multi-Label	One image may contain annotations from multiple classes simultaneously
Min Box Size	10×10 px; smaller defects skipped
No Merge	Overlapping defects of different classes each receive separate annotations
Partial Defects	Annotated if $\geq 50\%$ of defect is visible within frame

7. ANNOTATION EXECUTION

7.1 Annotated Defect Examples

Loose Thread & Press-Off (BBox + BBox): Red fabric sample with two simultaneous defects.

Press-off enclosed in tight yellow bounding box; loose-thread received a separate smaller box.

Demonstrates multi-label capability. *Observation:* Heavy patterns required 200% zoom in Roboflow for precise placement.

Loose Thread & Mis-Threaded (BBox + Polygon):

Beige/gold fabric showing mixed annotation types in one image — loose-thread in bounding box; misplaced/mis-threaded in smart polygon tracing the

weave disruption contour. Handled successfully by Roboflow’s layered annotation panel.

Fly Yarn & Snag (Multi-Instance): Dark patterned fabric with 7 total annotations — 4 fly yarn, 3 snag.

Each instance independently annotated with colour-coded class indicators. *Observation:*

Required careful discrimination between background pattern and actual defect regions.

Snag & Loose Thread (Multiple Instances):

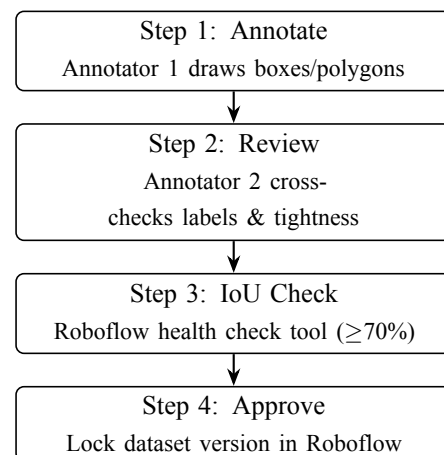
Grey/beige fabric with 2 snag and 2 loose-thread instances. Annotators referenced defect taxonomy guide to correctly distinguish visually similar classes.

7.2 Annotation Statistics

Metric	Value
Total images	846
Total annotation instances	$\approx 2,100+$
Single-label images	$\sim 40\%$
Multi-label images	$\sim 60\%$
BBox-annotated classes	12
Polygon-annotated classes	8
Min box size enforced	10×10 px
Max padding allowed	5 px

8. QUALITY ASSURANCE WORKFLOW

Quality Target: $\geq 70\%$ IoU agreement across annotators before locking dataset version.



Images scoring below 70% IoU are returned to Annotator 1 for correction and re-submitted through the full pipeline before approval.

9. DATASET PREPARATION FOR TRAINING

9.1 Dataset Split

Split	Prop.	Images	Purpose
Train	70%	≈592	Weight learning
Validation	20%	≈169	Hyperparameter tuning
Test	10%	≈85	Final evaluation

9.2 Data Augmentation Plan

Technique	Purpose
H/V Flip	Fabric orientation invariance
Rotation ($\pm 15^\circ$)	Handle varied camera angles
Brightness/Contrast	Simulate lighting variation
Gaussian Blur	Simulate motion blur
Mosaic	Increase multi-defect diversity
Random Crop & Resize	Scale invariance for small defects

9.3 Export Format

Exported from Roboflow in **YOLOv11 (PyTorch)** format:

- `images/train/`, `images/val/`, `images/test/`
- `labels/train/`, `labels/val/`, `labels/test/`
- `data.yaml` — class names and dataset path config

10. MILESTONE 2 – DELIVERABLES

1. **Annotated Dataset (Roboflow)** — 846 images across 20 classes; version locked after QA.
2. **Annotation Completion Report** — Per-class annotation counts and multi-label statistics.
3. **QA Validation Record** — IoU scores per batch; sub-70% images re-annotated and re-validated.
4. **Exported Dataset (YOLOv11)** — `train/val/test` image and label folders with `data.yaml`.
5. **Augmentation Plan** — Six augmentation strategies defined and implemented in Roboflow.

6. **Defect Visual Reference Guide** — Screenshot-based guide for all 20 defect classes.

7. **Challenges Log** — Nine annotation challenges and corresponding mitigation strategies documented.

11. CONCLUSION

The annotation and dataset preparation phase (Milestone 2) has been completed successfully. All 846 fabric defect images across 20 defect categories were annotated in Roboflow using a structured dual-annotator QA workflow with IoU-based acceptance criteria. Both bounding box and smart polygon strategies were applied appropriately based on defect morphology.

The annotated dataset has been exported in YOLOv11 format and is ready for model training. Challenges including visual similarity, complex backgrounds, and class imbalance were addressed through annotation discipline, reference guides, and a six-technique augmentation plan.

Next Steps — Milestone 3:

1. Train YOLOv11 on the exported dataset using Google Colab GPU.
 2. Evaluate precision, recall, `mAP@0.5` and `mAP@0.5:0.95` per class.
 3. Develop Python inference script for real-time fabric scan processing.
 4. Classify defects, record meter positions, write batch-level logs to CSV.
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